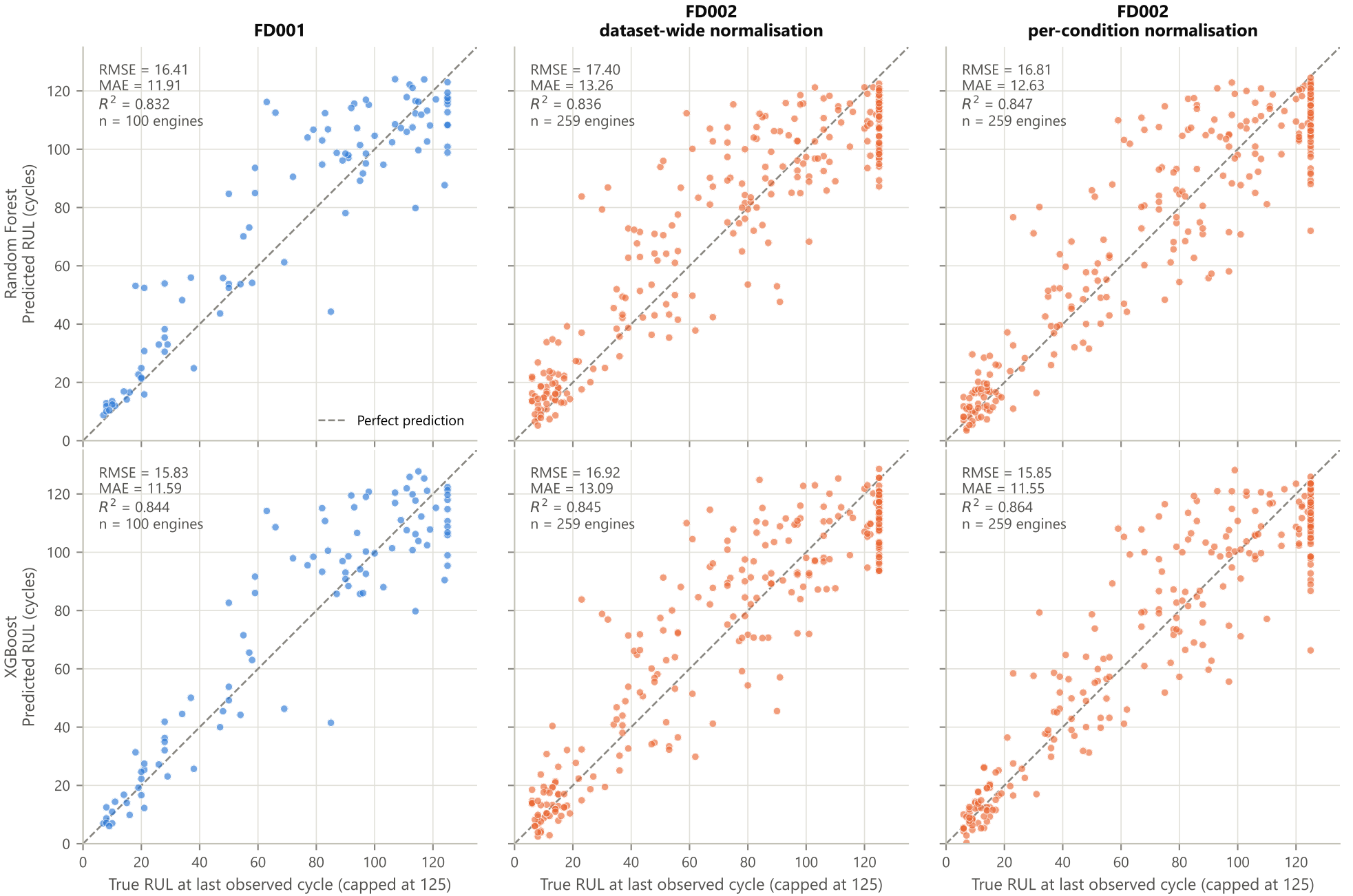
Actual vs predicted RUL on the official test engines (summary statistics)
FD001
FD002
dataset-wide normalisation
FD002
per-condition normalisation
RMSE = 16.41
MAE = 11.91
$R^2$ = 0.832
n = 100 engines
RMSE = 17.40
MAE = 13.26
$R^2$ = 0.836
n = 259 engines
RMSE = 16.81
MAE = 12.63
$R^2$ = 0.847
n = 259 engines
RMSE = 15.83
MAE = 11.59
$R^2$ = 0.844
n = 100 engines
RMSE = 16.92
MAE = 13.09
$R^2$ = 0.845
n = 259 engines
RMSE = 15.85
MAE = 11.55
$R^2$ = 0.864
n = 259 engines
Perfect prediction
Random Forest
Predicted RUL (cycles)
XGBoost
Predicted RUL (cycles)
0
20
40
60
80
100
120
True RUL at last observed cycle (capped at 125)